

NHID-Auth v1.0

Authorization Layer Specification

Companion Specification to NHID-Clinical v1.3

Delegated Authority Verification for Healthcare Voice AI

Author: Brianna Baynard

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Abstract

NHID-Auth addresses the authorization gap identified in NHID-Clinical v1.3: the ability to verify that an AI voice agent calling on behalf of a healthcare provider has legitimate delegated authority to access operational data. This companion specification defines a minimum viable attestation format for proof of delegation, separate from the behavioral disclosure requirements in NHID-Core. The authorization layer is intentionally decoupled to allow vendors to implement disclosure controls immediately while authorization infrastructure matures through real-world feedback.

1. The Authorization Problem

NHID-Clinical v1.3 Core requires AI voice agents to disclose their non-human identity before data exchange. However, disclosure alone does not solve the verification problem: How does the receiving party confirm the AI agent is authorized to represent the claimed provider?

Public identifiers like National Provider Identifier (NPI) do not prove authorization. Anyone can Google a provider's NPI and claim to call on their behalf. This creates an NPI spoofing vulnerability where unauthorized systems can extract protected operational data by naming a valid provider without proving delegation.

2. Core Principle: Identity $\neq$ Authorization

The fundamental distinction this specification makes:

Identity: Who or what is calling (answered by NHID-Clinical disclosure)

Authorization: Does the caller have permission to act on behalf of the named entity?

Public identifiers establish identity. Authorization requires proof of delegation.

3. Authorization Model Requirements

A valid authorization attestation MUST contain five elements:

Element	Description	Example
Delegating Entity	Who granted permission	Provider NPI or Payer Organization ID
Authorized Actor	System acting on behalf of delegating entity	Vendor platform ID or AI system identifier
Scope	What actions are permitted	['claims_inquiry', 'eligibility_check']
Duration	Validity period with start/end timestamps	2026-05-01T00:00:00Z to 2027-05-01T00:00:00Z
Revocation Capability	Mechanism for immediate invalidation	Reference ID for lookup or revocation endpoint

4. Authorization Attestation Requirements

A valid authorization attestation must contain five core elements. Organizations may implement these requirements using JSON, JWT, database records, or bilateral agreements — the format is not prescribed.

Required Elements:

Element	Purpose	Example Value
Delegating Entity	Who granted permission	Provider NPI or Payer Organization ID
Authorized Actor	System acting on behalf	Vendor platform ID or AI system identifier
Scope	Permitted actions	Claims inquiry, eligibility verification
Duration	Validity period	Start date through end date in ISO 8601 format
Revocation Capability	Invalidation mechanism	Reference ID or revocation endpoint

5. Validation Requirements

Receiving parties validating authorization should verify:

1. **Existence of Delegation:** Attestation exists and contains all five required elements
2. **Scope Limitation:** Requested action falls within the defined scope
3. **Temporal Validity:** Current timestamp falls within the validity period
4. **Revocation Awareness:** Attestation has not been revoked (if revocation infrastructure exists)

Organizations may validate attestations through internal databases, third-party registries, or bilateral verification agreements. NHID-Auth does not mandate a specific validation architecture.

6. Design Intent and Future Evolution

NHID-Auth v1.0 is intentionally minimal to enable immediate adoption while the ecosystem develops verification infrastructure. Key design principles:

Portable: No cryptographic infrastructure required at baseline

Audit-Friendly: Human-readable format for compliance review

Extensible: Upgrade path to JWT, PKI, or registry-based systems

Decoupled: Authorization layer independent of NHID-Core behavioral requirements

Future Evolution (v1.1+):

- JWT-based attestations with signature verification
- Central registry for real-time authorization lookup
- Scope negotiation protocols
- Revocation APIs and certificate revocation lists
- Integration with FHIR authorization frameworks

7. Relationship to NHID-Clinical v1.3 Core

NHID-Auth is a companion specification, not a replacement. The two layers work together:

Layer	Addresses	Status
NHID-Core v1.3	Behavioral disclosure (who is calling, when they call, etc.)	Stable, implementable
NHID-Auth v1.0	Authorization verification (proof of delegation to a third party)	Baseline infrastructure maturing

Organizations can implement NHID-Core behavioral controls immediately and adopt NHID-Auth authorization verification when organizational partnerships and verification infrastructure are ready.

8. Implementation Guidance

Vendor Implementation:

1. Generate attestation with delegating_entity from provider contract
2. Include reference_id for audit trail
3. Set scope to minimum actions required (principle of least privilege)
4. Implement expiration and renewal workflow
5. Provide revocation endpoint or invalidation mechanism

Payer/Receiver Implementation:

1. Request authorization attestation during call
2. Validate attestation structure and required fields
3. Check scope matches requested action
4. Verify temporal validity (issued_at, expires_at)
5. Log reference_id for audit purposes

9. References

- NHID-Clinical v1.3 Core Specification
- OAuth 2.0 Authorization Framework (RFC 6749)
- JSON Web Token (JWT) Specification (RFC 7519)
- NIST Special Publication 800-63-3: Digital Identity Guidelines
- HIPAA Security Rule: Access Control Standards (45 CFR §164.308(a)(4))

10. License and Attribution

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11. Contact Information

Technical Questions: help@nhid-clinical.org

Website: nhid-clinical.org

Community: Discord (discord.gg/eP8FxXkGU6) | Reddit ([r/NonHumanAuth](https://www.reddit.com/r/NonHumanAuth))